Societal impacts of AI

Ethics and Bias in AI: week 2

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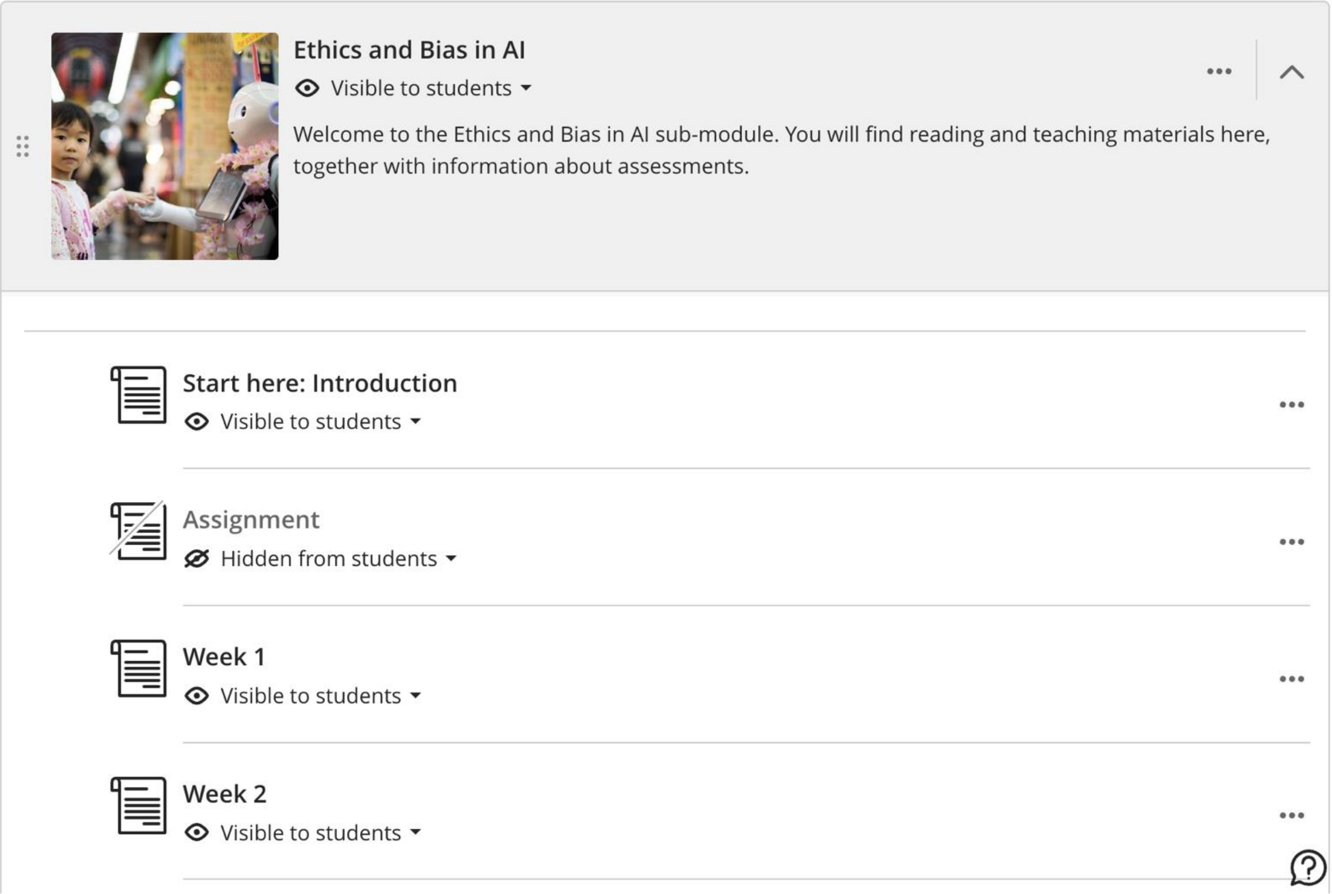
House keeping

- Office hours
  - MSC10029
  - Friday 11.00-12
  - Email to arrange
- Practicals
  - Start next week
- Blackboard Ultra
  - Reading and tasks

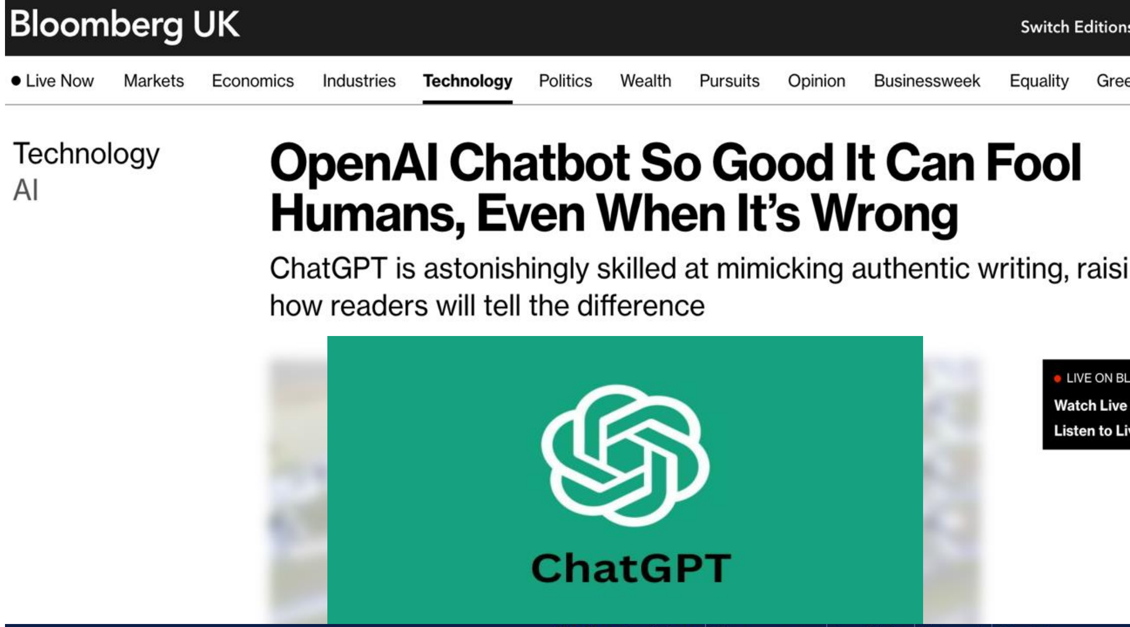


Last lecture…

Risks, harms & moral implications

Risks, harms & moral implications

- Environmental impacts
- Value alignment issues
- Harmful content, inaccuracies
- Privacy issues, data breaches
- Exploitation of workers
- Copyright infringement
- Labour force / sector transformation
- Unequal access to AI as costs evolve
- Impacts on social and individual wellbeing



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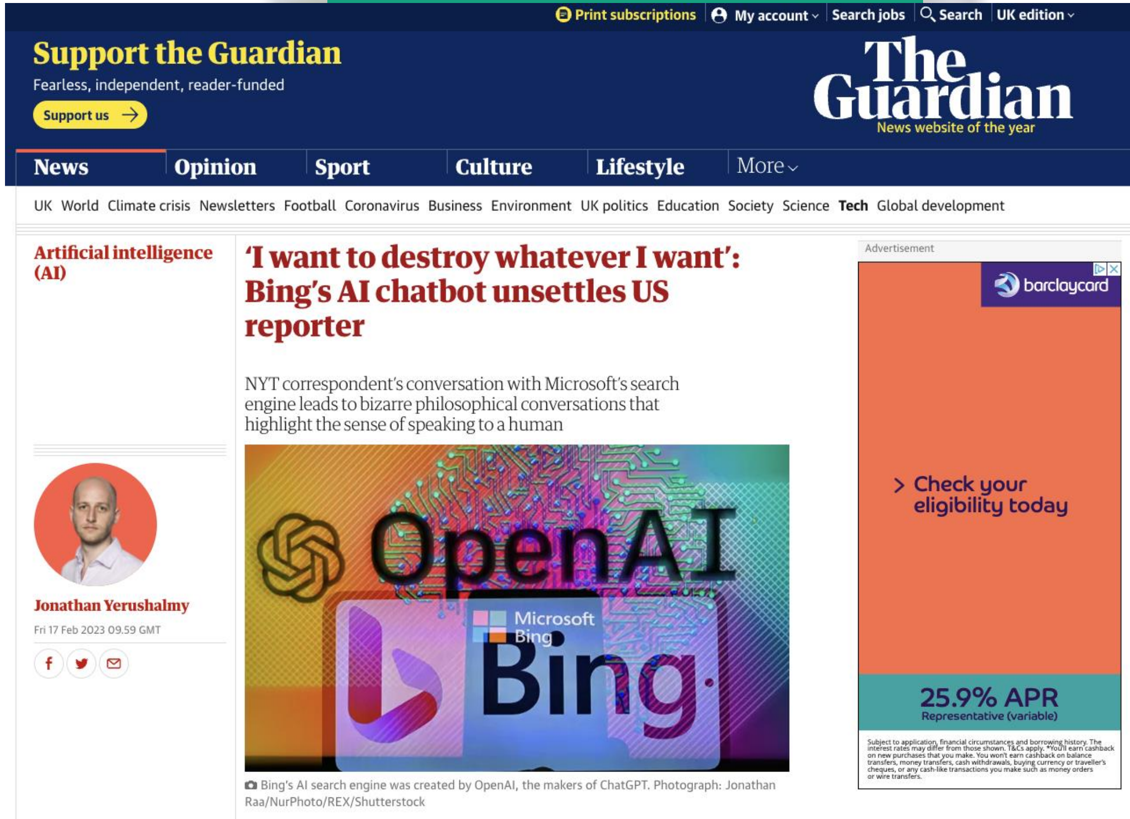
Technology
AI

OpenAI Chatbot So Good It Can Fool Humans, Even When It's Wrong

ChatGPT is astonishingly skilled at mimicking authentic writing, raisi
how readers will tell the difference



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Artificial intelligence (AI)

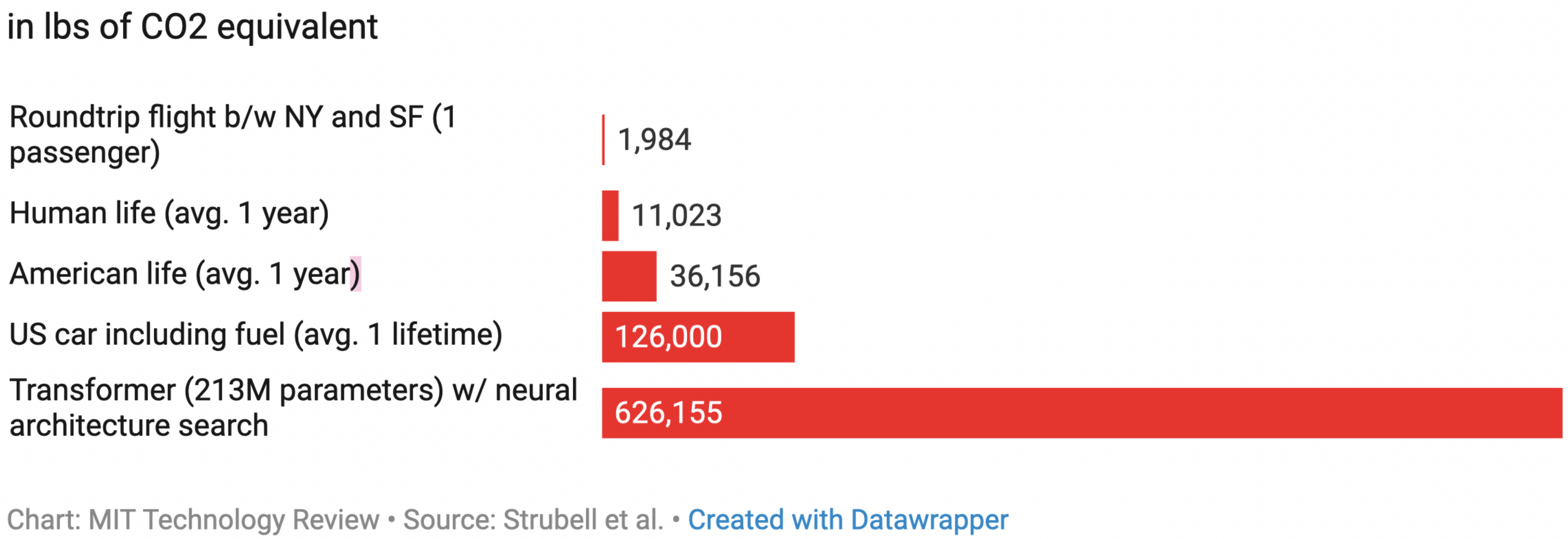
'I want to destroy whatever I want': Bing's AI chatbot unsettles US reporter

NYT correspondent's conversation with Microsoft's search engine leads to bizarre philosophical conversations that highlight the sense of speaking to a human

Jonathan Yerushalmy
Fri 17 Feb 2023 09.59 GMT

Bing's AI search engine was created by OpenAI, the makers of ChatGPT. Photograph: Jonathan Raa/NurPhoto/REX/Shutterstock

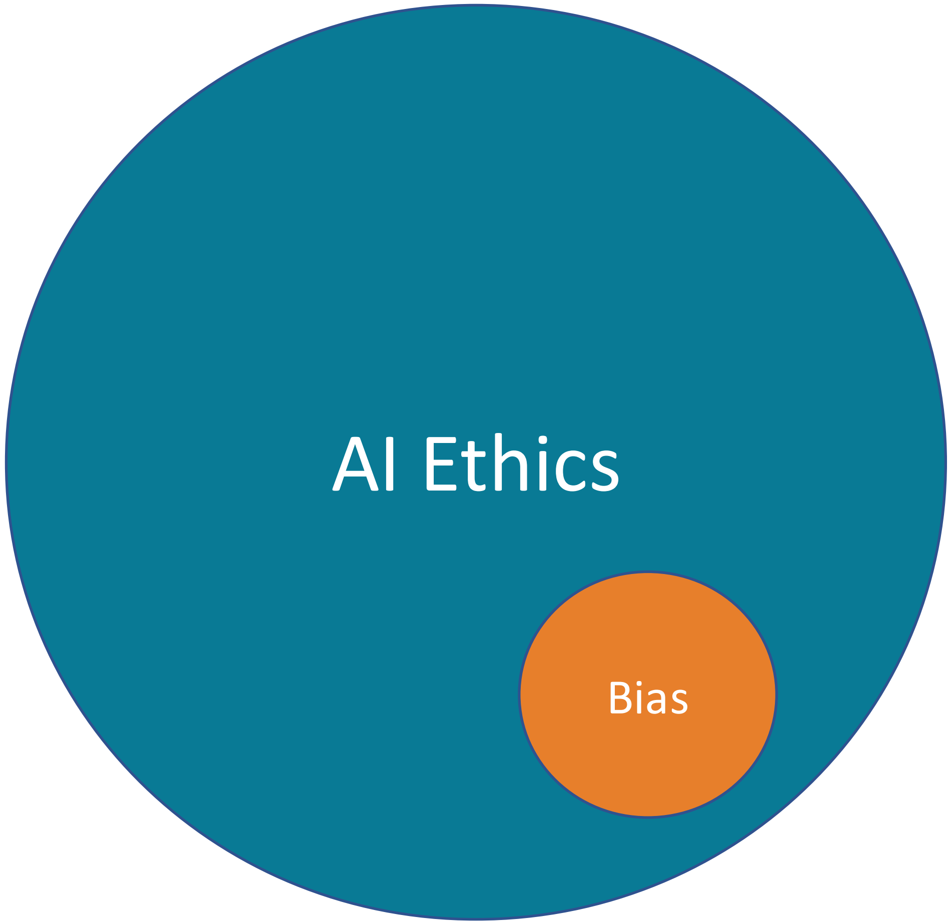
Common Carbon Footprint Benchmarks



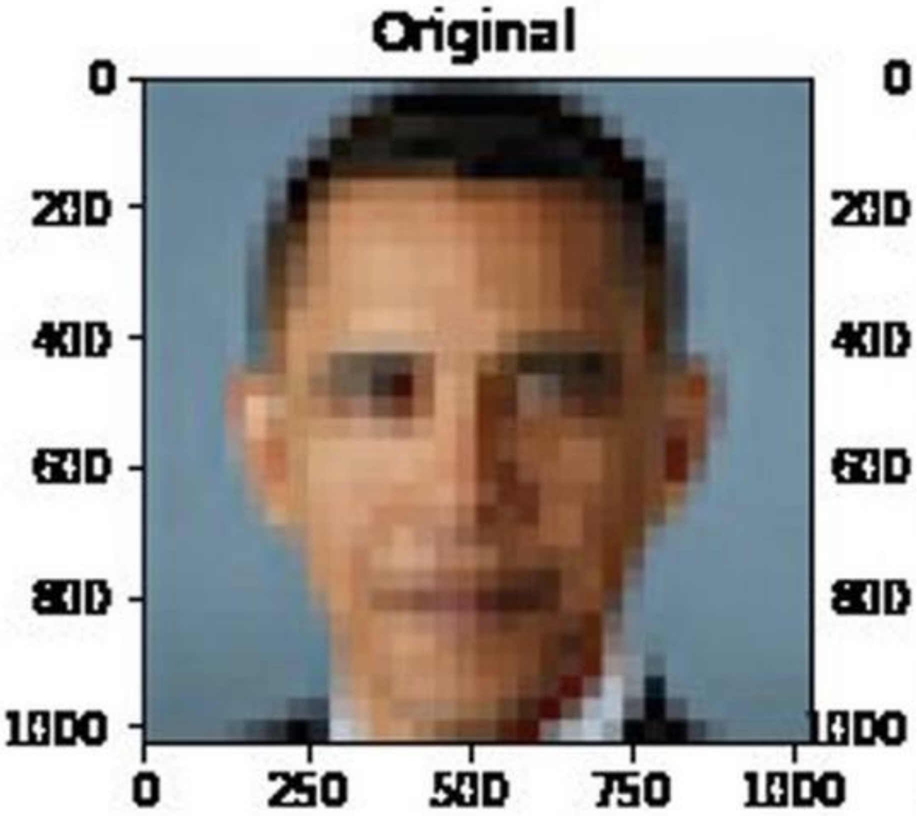
Lecture overview

- What are biases in human thought?
- What are the implications of human biases?
- What is bias in AI?
- Examples of algorithmic bias and their consequences

What is bias?

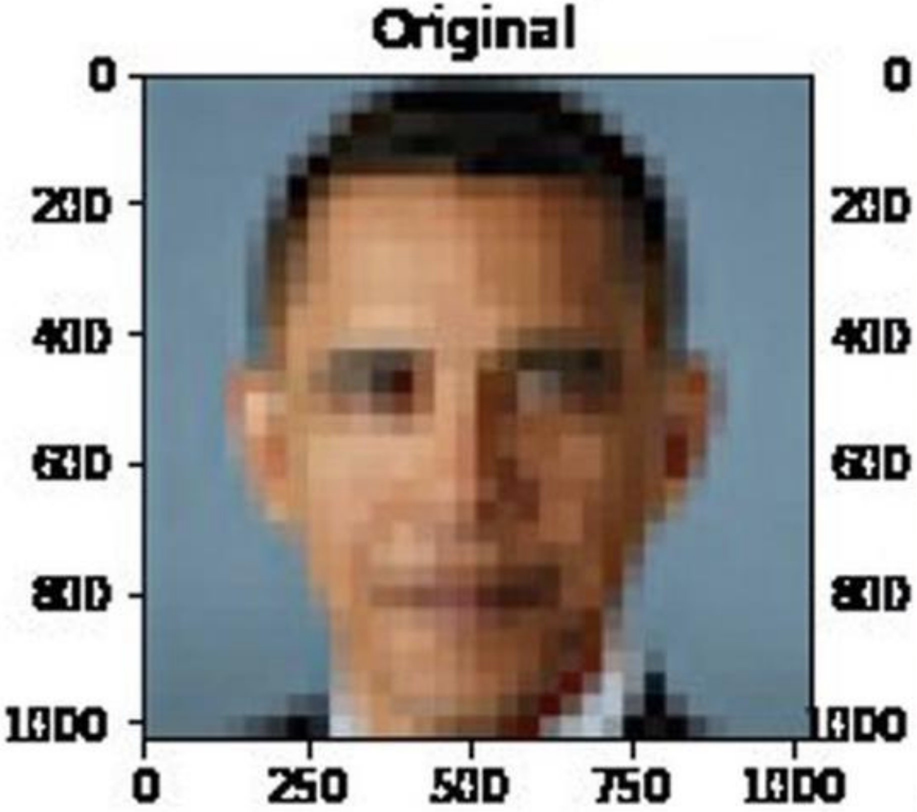


Algorithmic bias?



Input to the PULSE face depixeliser.

Algorithmic bias?



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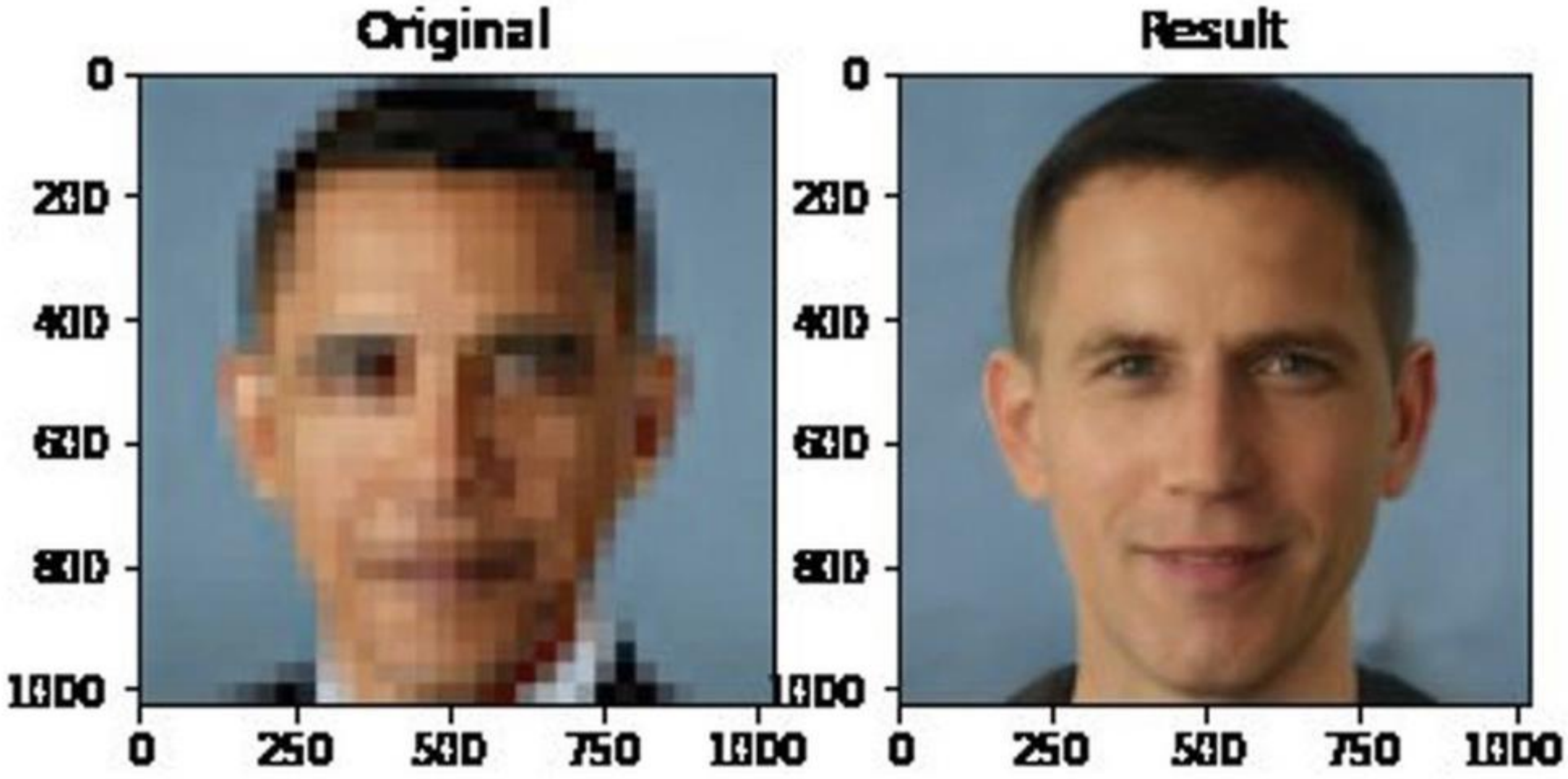
What is bias?



Output of the PULSE face depixeliser. The underlying image is of Barrack Obama, but the output from the model, informed by biased training data, outputs a caucasian face.

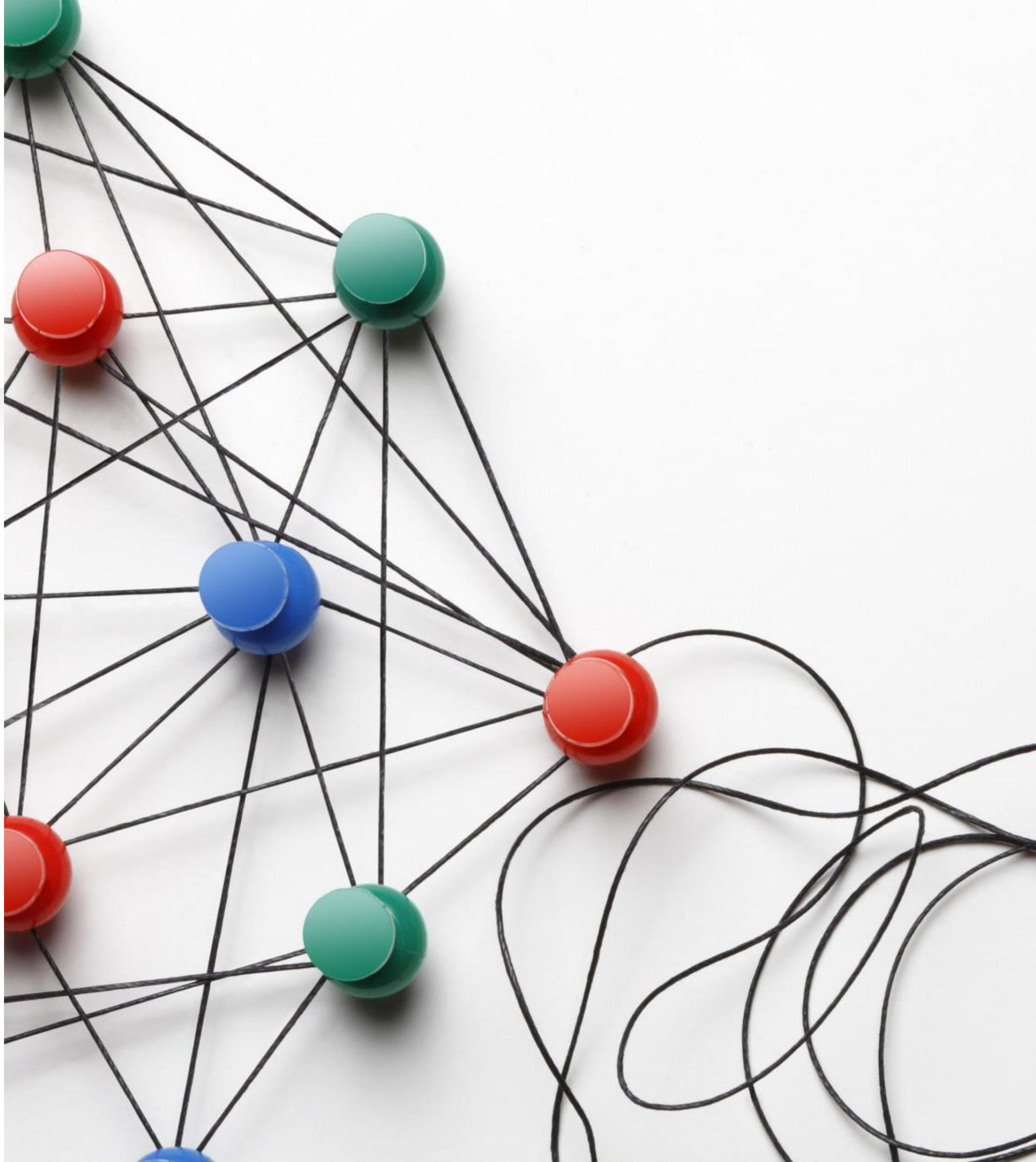
In statistics / machine learning…

In machine learning, bias refers to any basis for choosing one generalization over another, other than strict consistency with the instances (Mitchell, 1980).

The more the bias of a certain learning algorithm fits the characteristics of a particular problem, the better it will perform on that problem (Mooney, 1996)

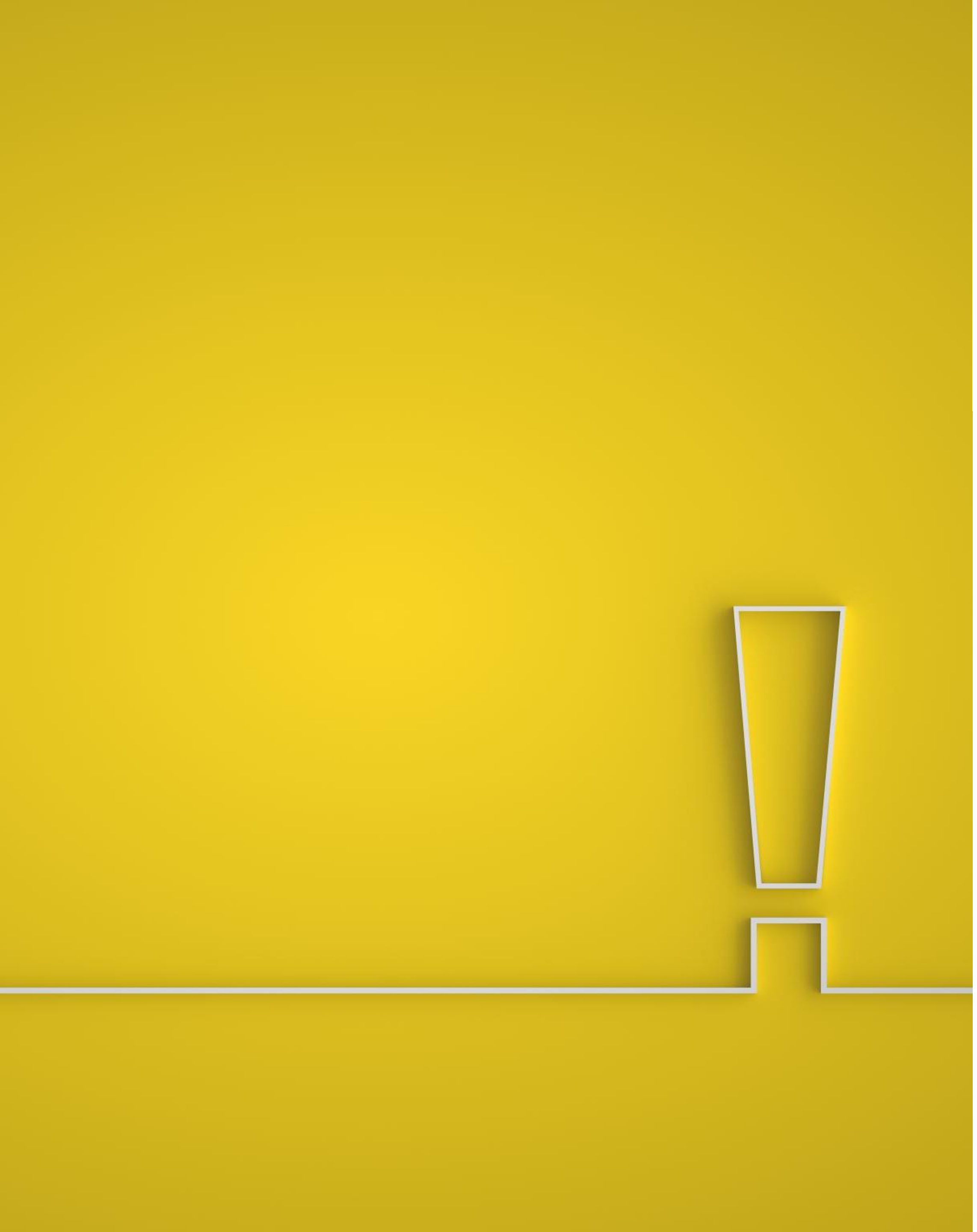
What is Bias – in humans?

- Simplify information processing through a filter of prior experience, preference.
- Help to make sense of the word and provide shortcuts for decision making.
- Not inherently bad, but can have negative consequences. Especially cognitive biases.
- Can contribute to perpetuating systemic inequalities.



Biases in human thought and judgement

- **Social biases:** where we automatically form impressions of people, or leap to conclusions, based on the social group that they are a member of.
  - E.g. instantly warm to someone who speaks with the same accent as us, or if we assume someone from a different ethnic group is unlikely to be telling the truth.
- **Cognitive biases:** systematic tendencies in our thought processes that can lead us into error.
  - E.g. confirmation bias, whereby we seek information which can confirm our beliefs, inadequately testing beliefs by seeking out potentially contradictory information.



Impacts on decision making and judgement

Exposure effect:

> “A reliable way to make people believe in falsehoods is frequent repetition, because familiarity is not easily distinguished from truth. Authoritarian institutions and marketers have always known this fact.”
>
> Daniel Kahneman,
> Thinking, Fast and Slow

Implicit biases

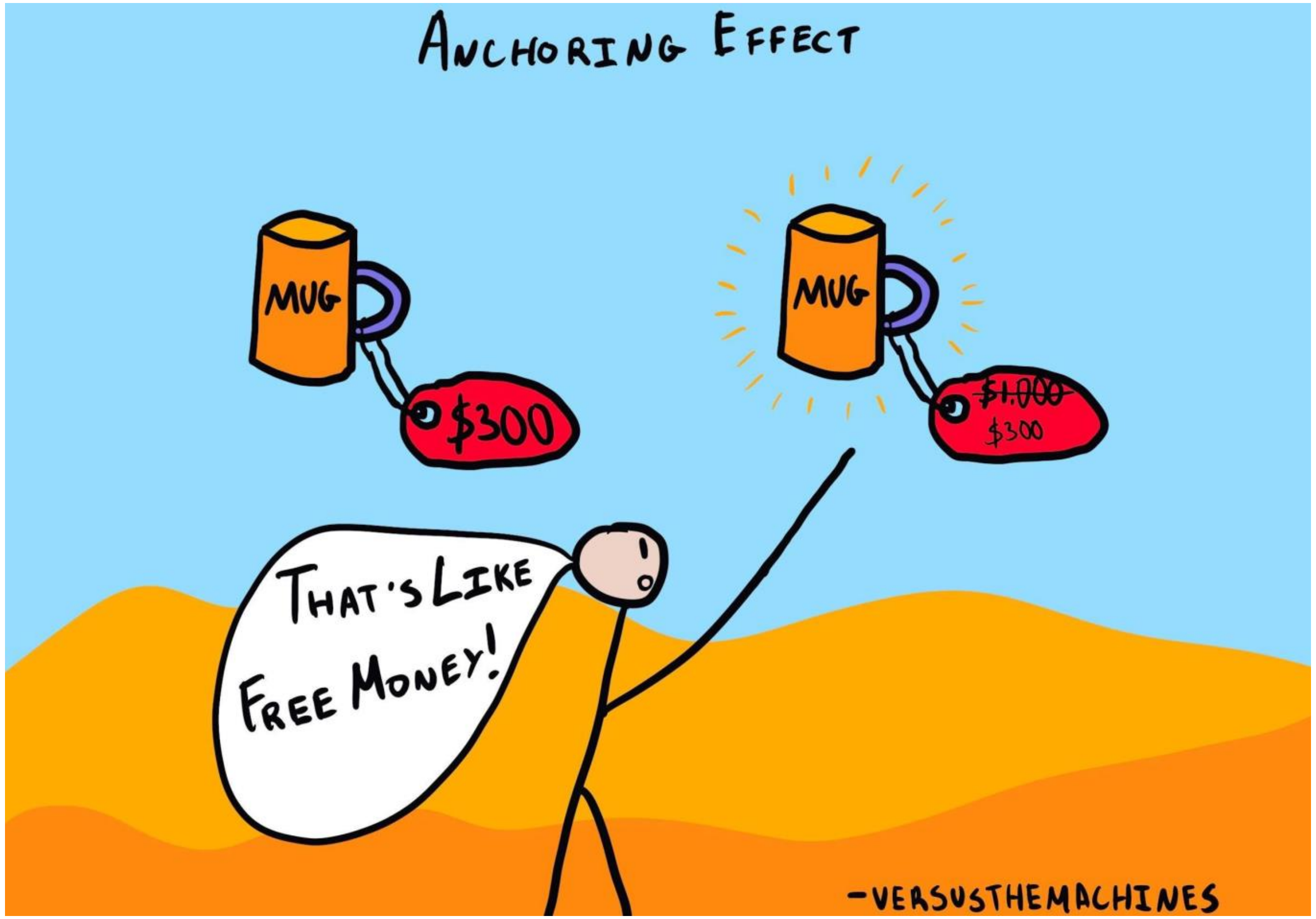
- Implicit biases: automatic associations which operate without reflective control - can lead to unintentionally differential or unfair treatment of stigmatised individuals.

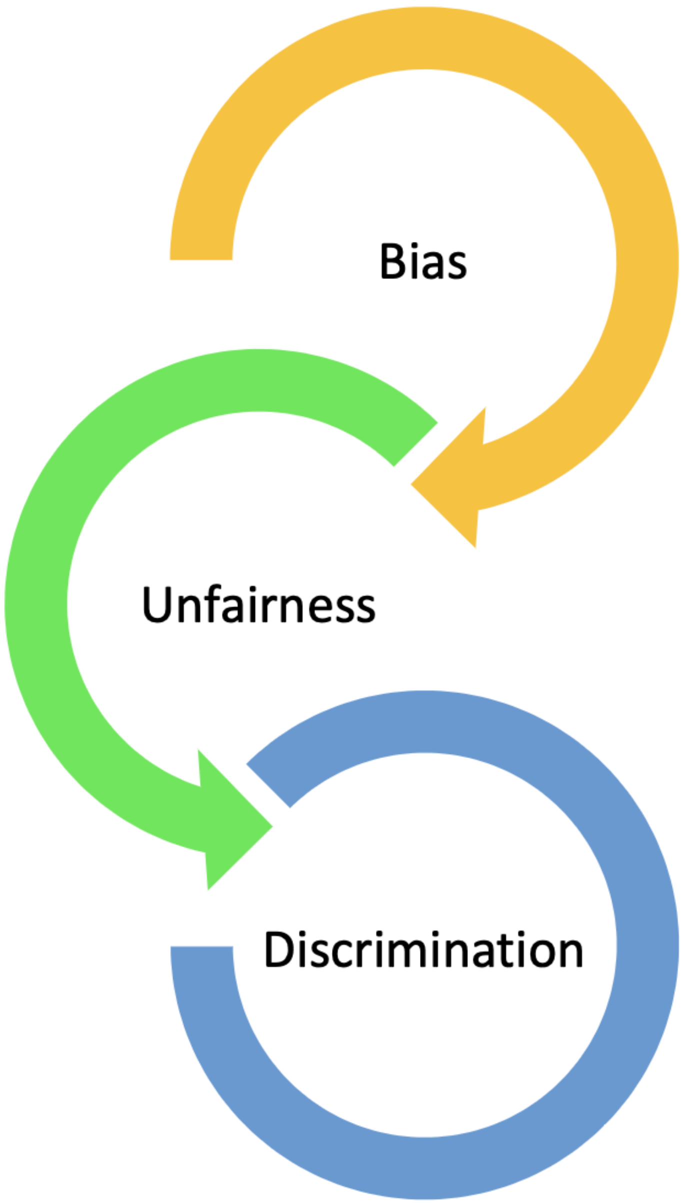
- Implicit Association Test

You can take a test to explore your own implicit biases:

https://implicit.harvard.edu/implicit/selectatest.html

Biases affect human decision-making





Bias & discrimination

From discriminate – ability to discern a difference

Discriminate (Cambridge Dictionary):

to treat a person or particular group of people differently, especially in a worse way from the way in which you treat other people, because of their race, gender, sexuality, etc.

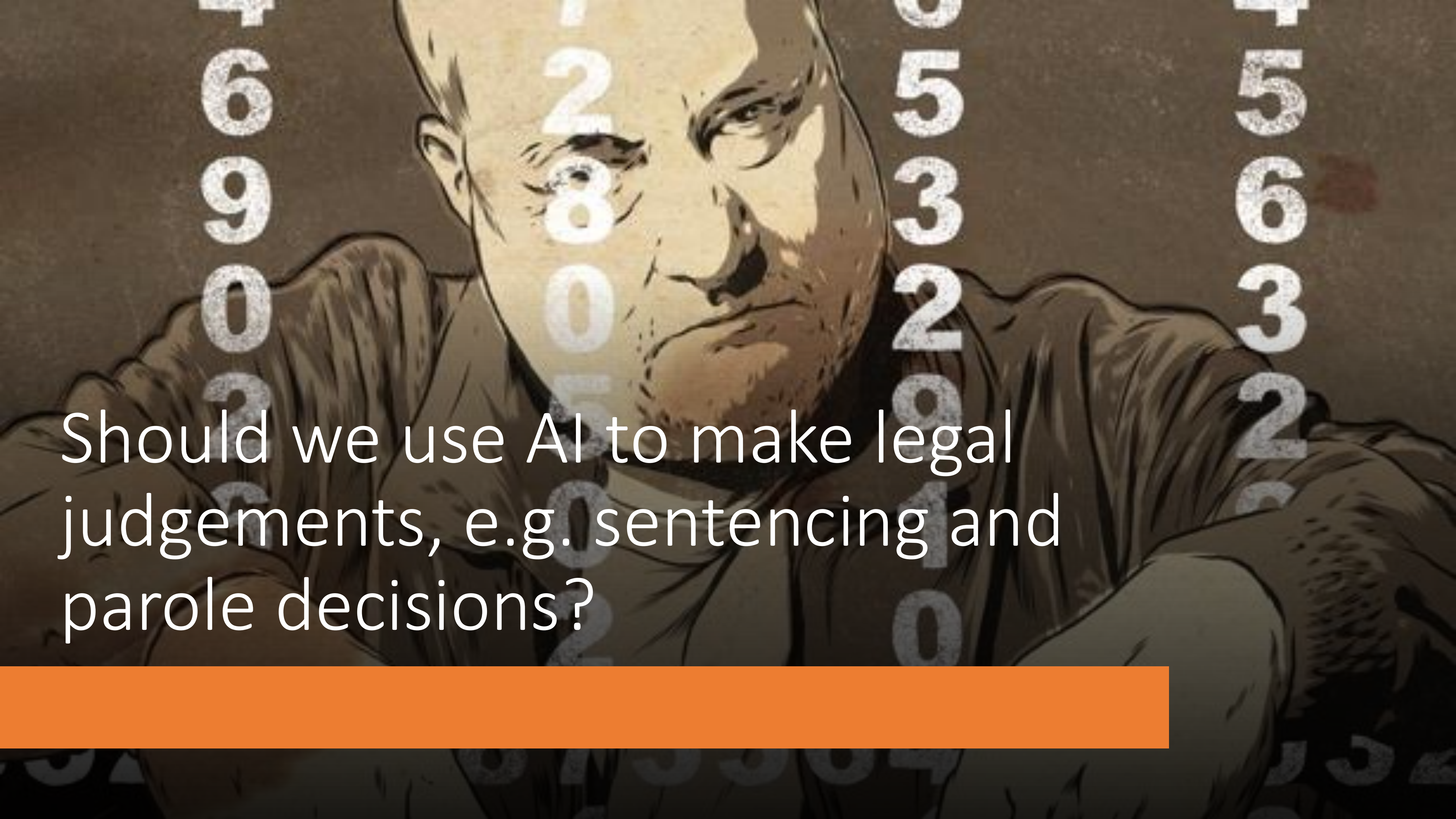
- Related concepts:
    - Discrimination
    - Bias
    - (Un)fairness

Attributes Associated with Social Bias

Certain individual attributes are tied to social bias (often referred to as ‘protected attributes’ or ‘characteristics’).

Laws that prohibit discrimination based on these attributes.
E.g. Equality Act 2010 establishes and defines 9 protected characteristics:

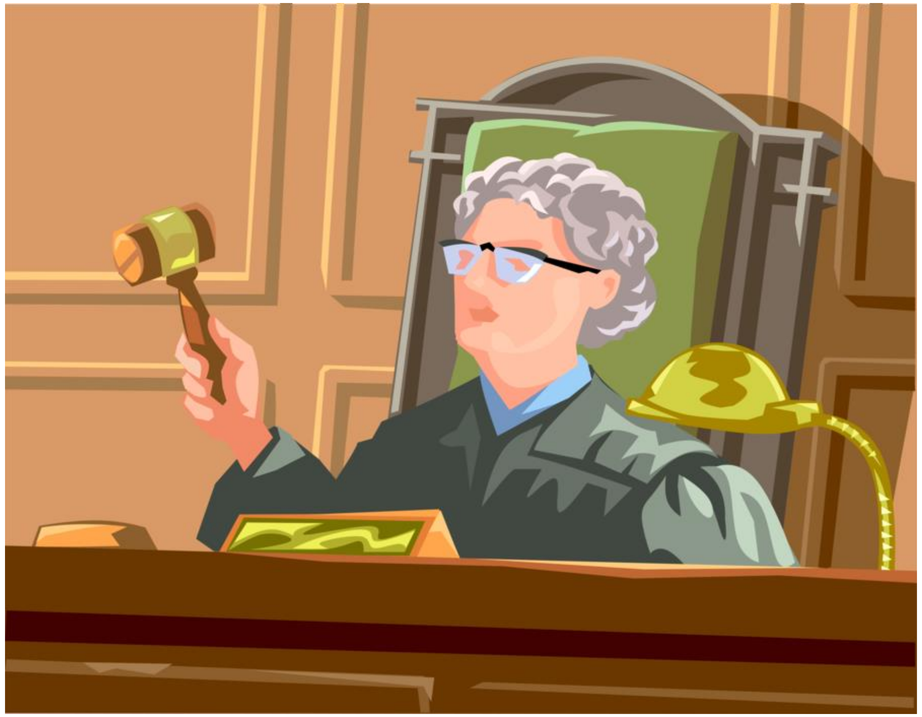
- Age
- Race
- Sex
- Sexual orientation
- Disability
- Marriage and civil partnership
- Gender reassignment
- Religion or belief
- Maternity and pregnancy



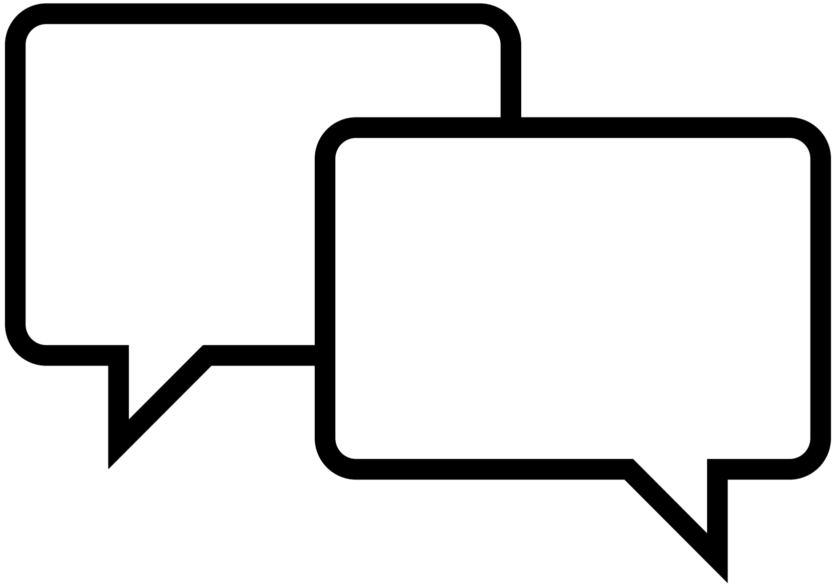
Should we use AI to make legal judgements, e.g. sentencing and parole decisions?

Case study: criminal justice algorithms

- The COMPAS algorithm (Correctional Offender Management Profiling for Alternative Sanctions)
- A computer algorithm used by judges across the US to help make decisions about bail, sentencing, and parole
- Draws on a range of data about defendants
- Produces a "risk score" from 1-10 predicting how likely someone is to reoffend
- Used to decide whether defendants should be detained or released on bail pending trial



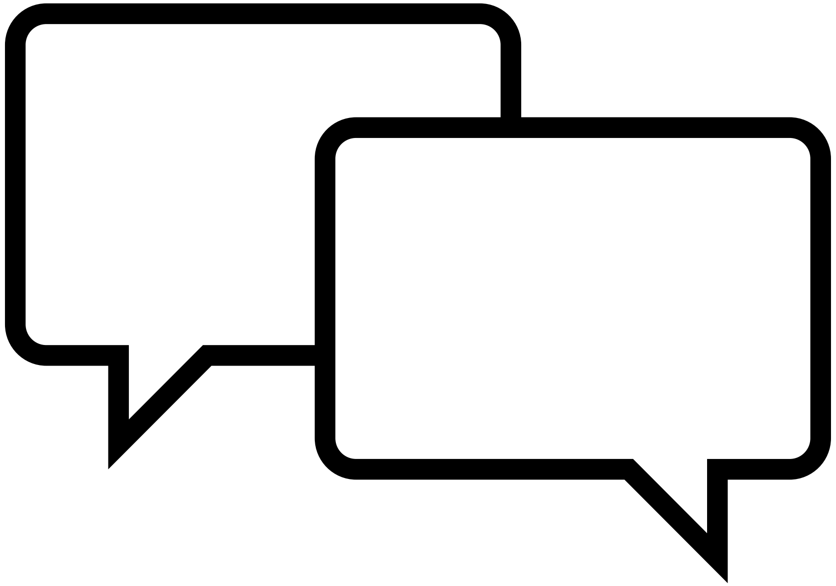
Discuss in groups (10 minutes)



Imagine you're designing this algorithm. You want to predict who *might* commit another crime.

What information would you want to know about someone to make that prediction?

Feedback on discussions



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What Data Does COMPAS Use?

Criminal History

- Number and types of prior arrests and convictions
- Age at first arrest
- Violence in criminal history

Demographic : Personal & Social Factors

- Current age
- Employment status and history
- Educational background
- Marital status and family situation
- Substance abuse history

Social Environment

- Number of friends/associates who have been arrested
- Social isolation measures
- Community ties and support systems

Attitudes & Personality

- Responses to questions about anger management
- Attitudes toward crime and punishment
- Self-reported impulsiveness
- Cooperation with supervision

COMPAS produces scores for:

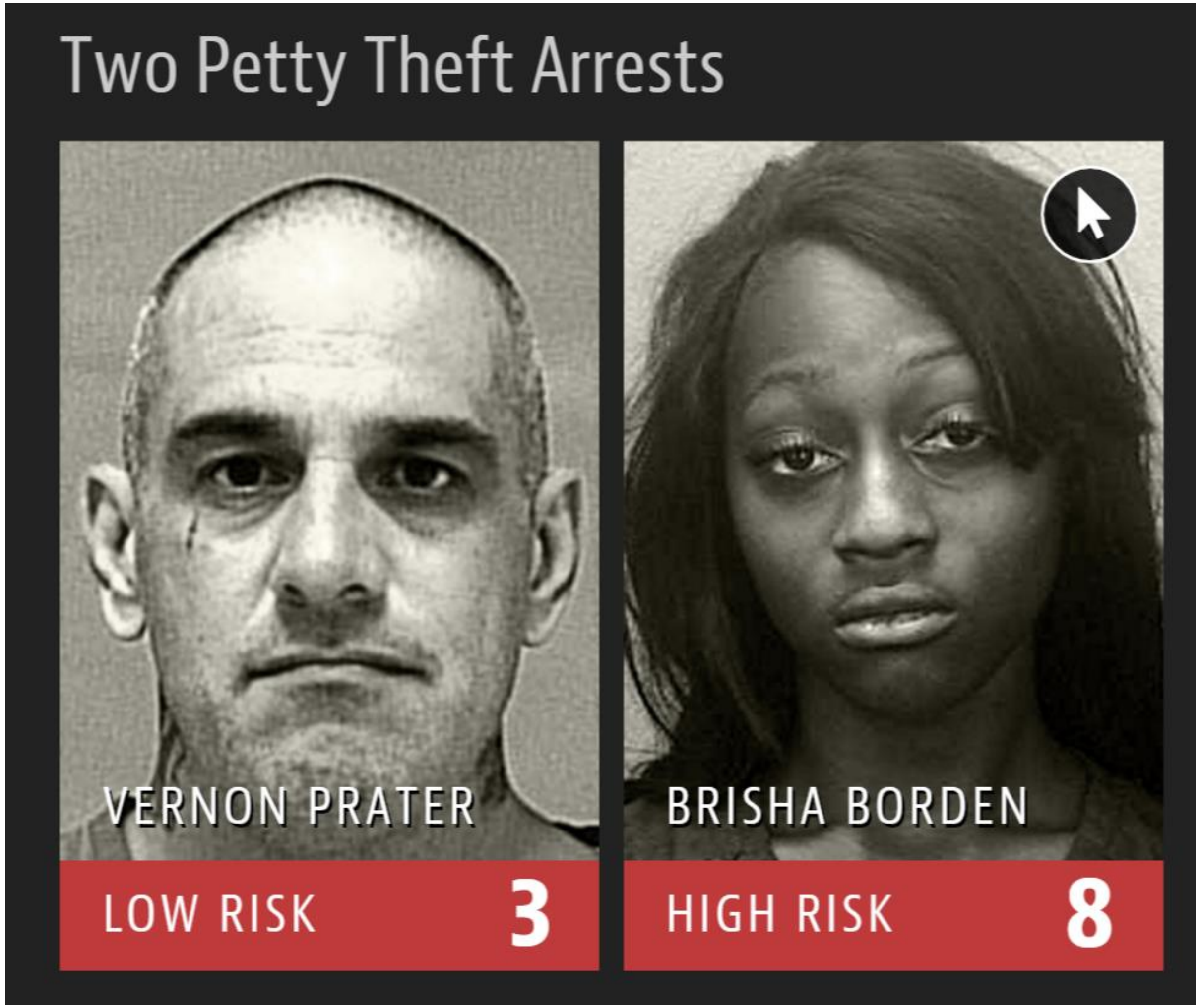
- General recidivism risk (1-10 scale)
- Violent recidivism risk (1-10 scale)
- Failure to appear risk (1-10 scale)

Algorithmic Bias

Shoplifted $86.35 worth of tools from Home Depot.

8 prior arrests.

Convicted of armed robbery and attempted armed robbery, for which he served five years in prison, in addition to another armed robbery charge.

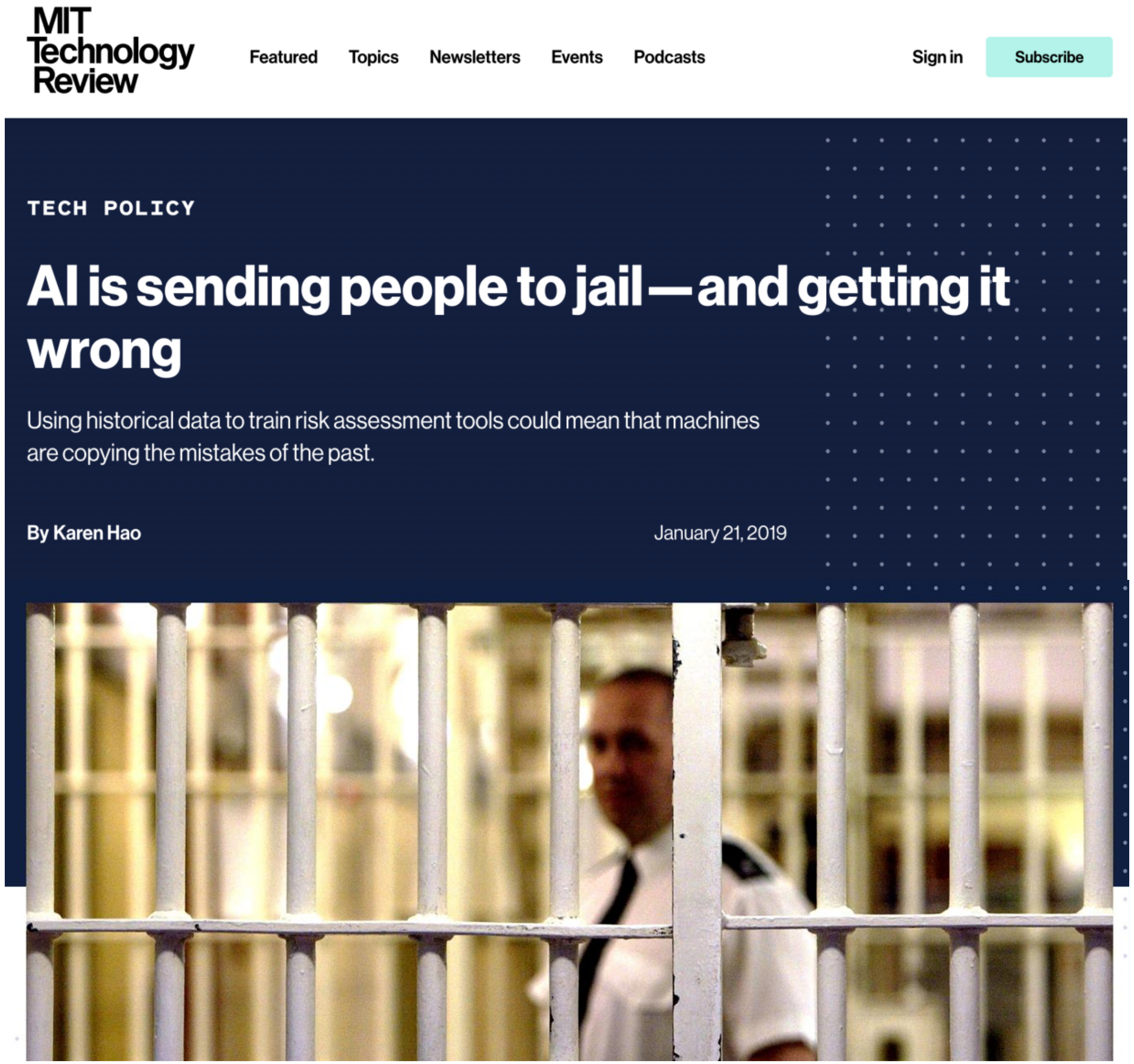


Took a child's bike and scooter. Value $80. Dropped them and ran.

4 Misdemeanors while a juvenile.

Got it backward!

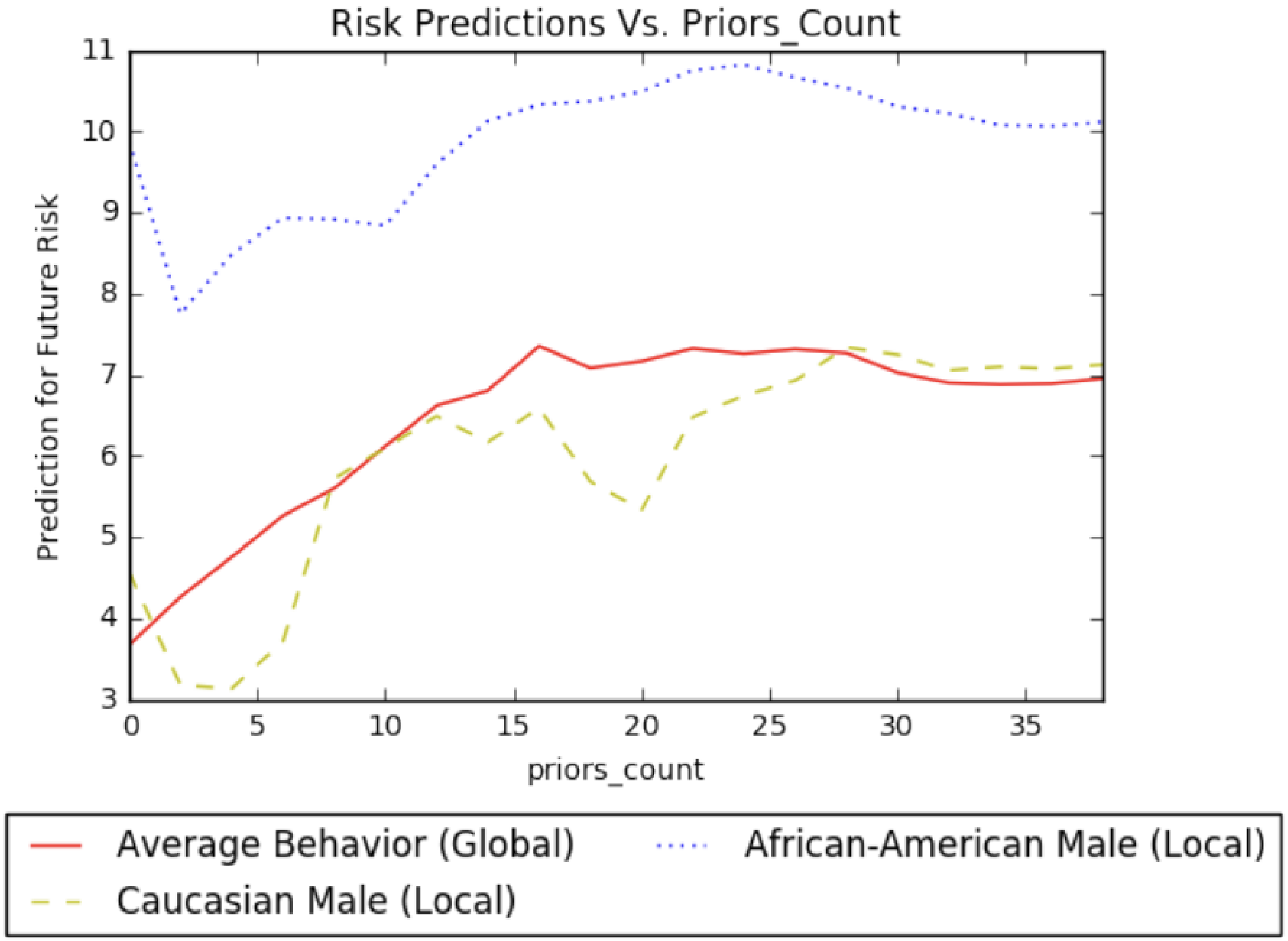
Bias in criminal justice algorithms



- African-Americans were more likely to be assigned a higher-risk score, resulting in longer periods of detention, compared to whites who were equally likely to re-offend.

Angwin, Julia, Jeff Larson, Surya Mattu, and Laura Kirchner. "Machine Bias." ProPublica, May 23, 2016. https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing

Bias in predicting risk of reoffending



- Northpointe, the firm that sells the algorithm's outputs, argues that wrong metrics are being used to assess fairness in the product.
- **But how should fairness be defined and measured?**

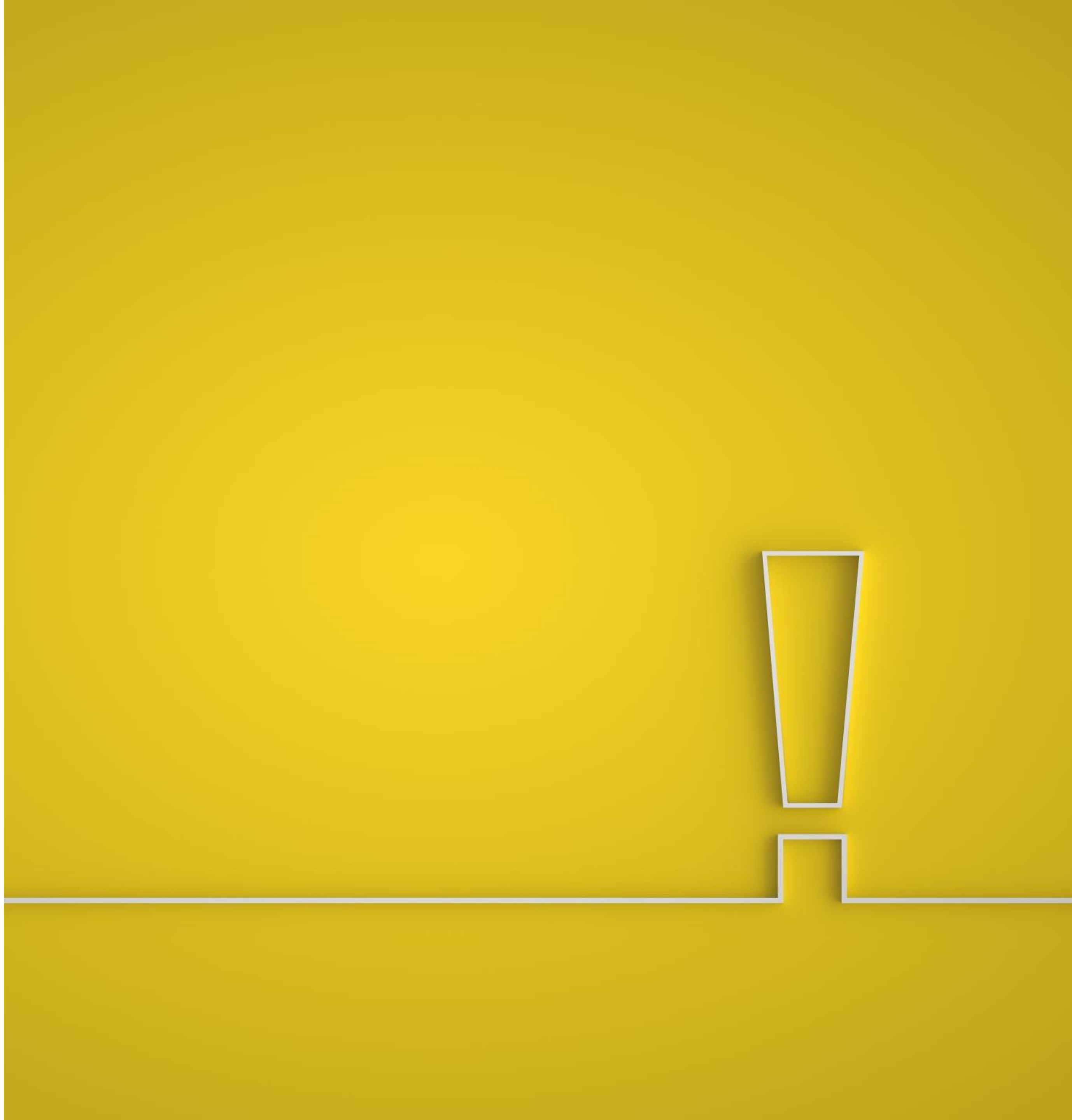
From: Hall, P., & Gill, N. (2017). Debugging the Black-Box COMPAS Risk Assessment Instrument to Diagnose and Remediate Bias.

Why would courts use it?

- Judges make these high-stakes decisions quickly with limited information
- Claims to be more "objective" than human judgment alone
- Supposed to reduce inconsistencies between different judges
- Can help identify low-risk offenders who might not need jail time

Question?

- If humans are flawed decision-makers does it matter if the algorithms are flawed too? If so, why?
- Is it better to use an algorithm if it is shown to systematically discriminate, but discriminates less than the human counterparts?
- What should be used to judge whether a system is adequate for use?



Stakeholder Perspectives

Stakeholder roles:

- Defendant
- Judge
- Victim's family
- General member of public
- Data scientist who built the system

In your groups discuss, from your stakeholder’s perspective, what would be your biggest concern about this AI system?

Some considerations

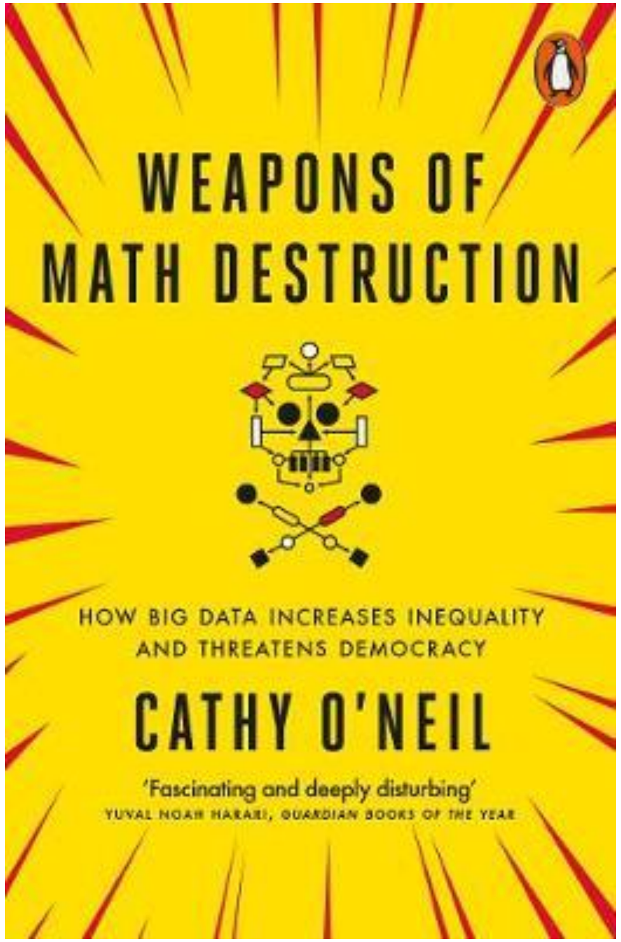
- People are more likely to assume algorithms are objective and error-free
  - Cathy O'Neil: The privileged are processed by people, the poor are processed by algorithms
- More likely to implement no appeal process
- Algorithmic systems are cheaper + more scaleable
- Machine Learning can amplify bias
- Machine Learning can create feedback loops
  - Feedback loops occur when your model is controlling the next round of data you get
  - E.g. predictive policing, youtube recommendation system

Technology is power, and with that, comes responsibility!

Cathy O'Neil, Weapons of Math Destruction (2016)

Mathematical models can "encod[e] human prejudice, misunderstanding, and bias into the software systems that increasingly manag[e] our lives. [...] Their verdicts, even when wrong and harmful, [are] beyond dispute or appeal. And they tend to punish the poor and the oppressed in our society, while making the rich richer"

(O'Neil, 2016, p. 3).

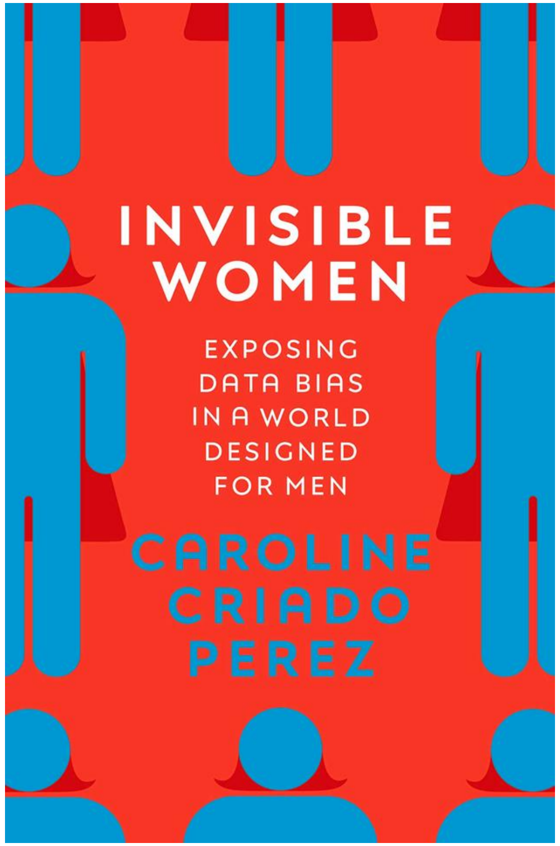


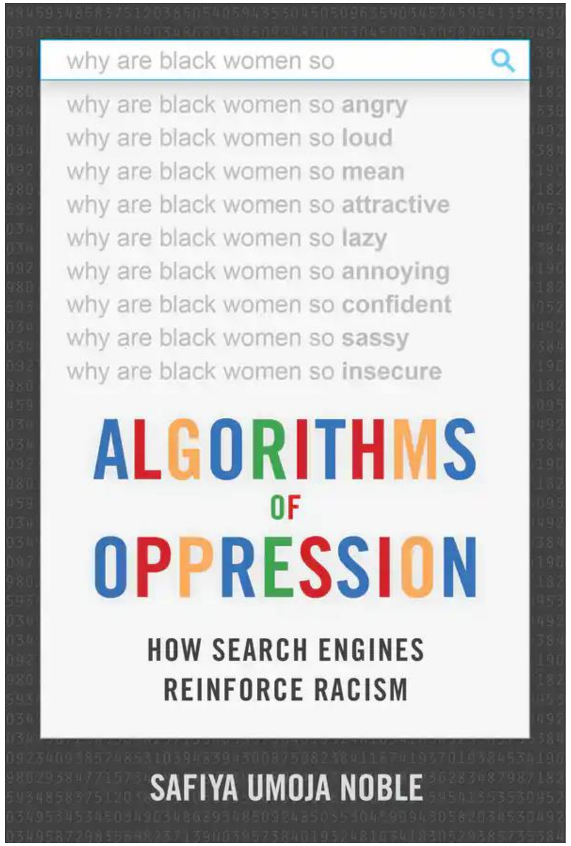
Friedman & Nissenbaum (1996).

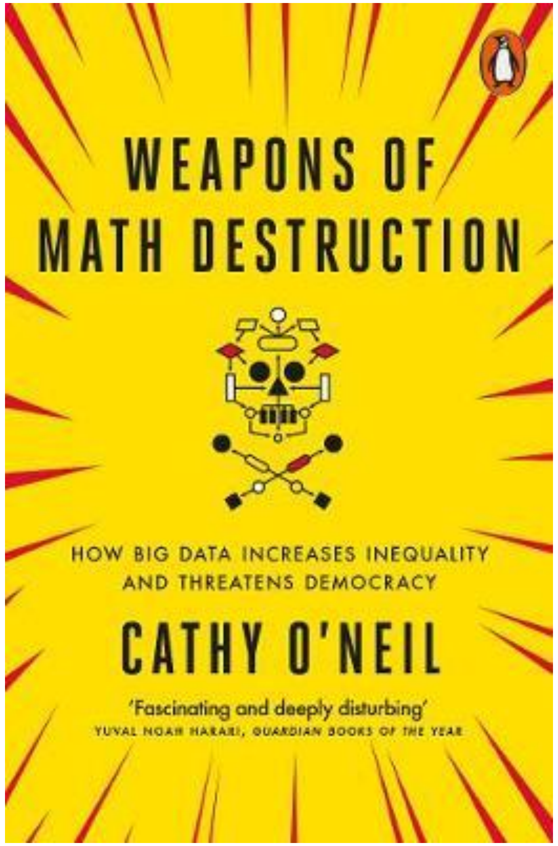
“Because biased computer systems are instruments of injustice—though admittedly, their degree of seriousness can vary considerably—we believe that **freedom from bias** should be counted among the select **set of criteria** according to which the **quality of systems** in use in society **should be judged**”.

What criteria?

What should be included in the “set of criteria according to which the quality of systems in use in society should be judged”

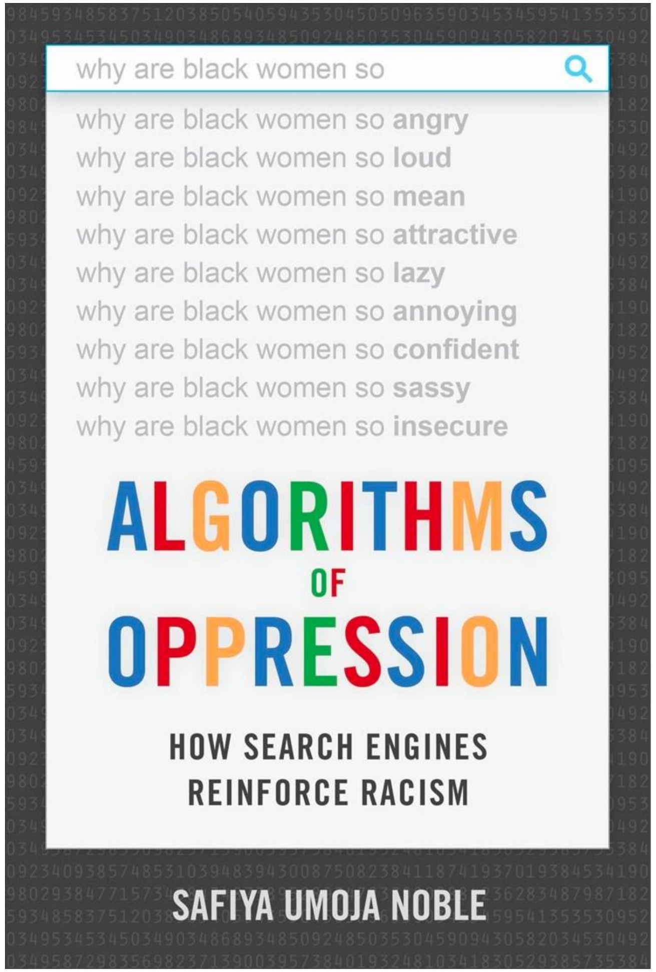






Bias: from data to algorithms and beyond

Bias in search and recommendation



- Safiya Noble (2018)
- An analysis of textual and media searches and paid online advertising
- Search results returned for "black girls" vs "white girls" were notably different
- Sexually explicit content far more common in the former
- "algorithmic oppression is not just a glitch in the system but, rather, is fundamental to the operating system of the web"

Bias in healthcare

https://www.theguardian.com/society/2021/nov/09/ai-skin-cancer-diagnoses-risk-being-less-accurate-for-dark-skin-study

https://www.theguardian.com/society/2021/nov/21/from-oximeters-to-ai-where-bias-in-medical-devices-may-lurk

The Guardian
News website of the year

NHS

This article is more than **1 year old**

Analysis

From oximeters to AI, where bias in medical devices may lurk

Nicola Davis
Science correspondent

Analysis: issues with some gadgets could contribute to poorer outcomes for women and people of colour

Sun 21 Nov 2021 17.56 GMT

This article is more than **1 year old**

AI skin cancer diagnoses risk being less accurate for dark skin – study

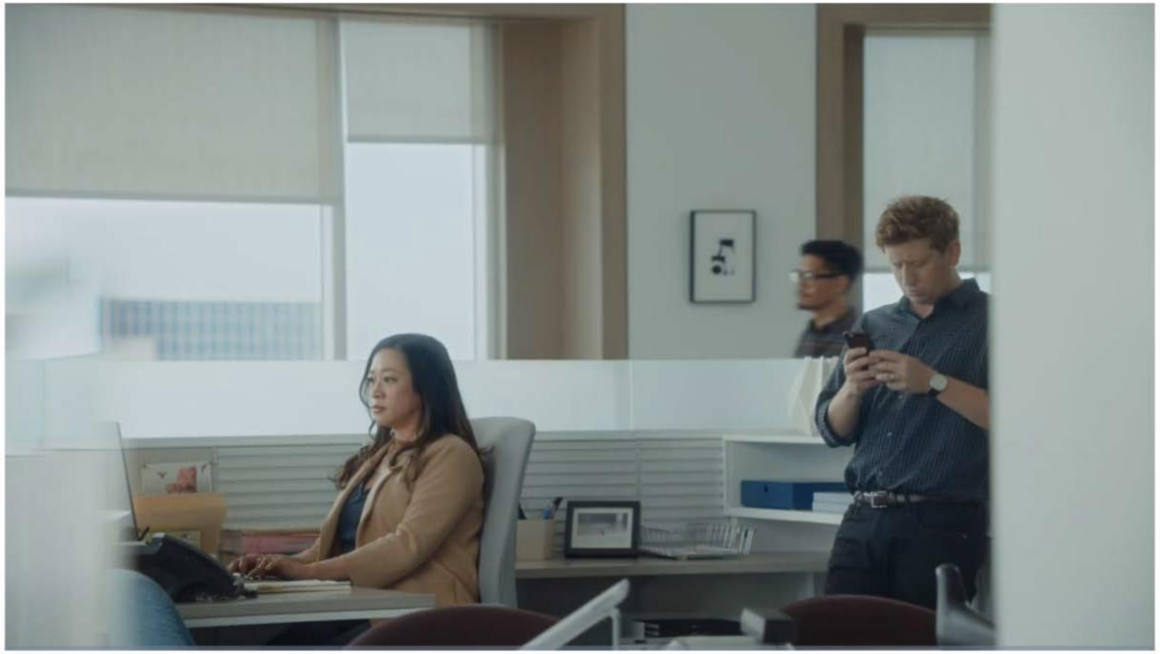
Research finds few image databases available to develop technology contain details on ethnicity or skin type

Studies suggest image recognition technology can classify skin cancers as successfully as humans. Posed by model. Photograph: ChesiireCat/Getty Images/iStockphoto

AI systems being developed to diagnose skin cancer run the risk of being less

MPW • AMAZON

Amazon Reportedly Killed an AI Recruitment System Because It Couldn't Stop the Tool from Discriminating Against Women



By DAVID MEYER October 10, 2018

Machine learning, one of the core techniques in the field of artificial intelligence, involves teaching automated systems to devise new ways of doing things, by feeding them reams of data about the subject at hand. One of the big fears here is that biases in that data will simply be reinforced in the AI systems —and Amazon seems to have just provided an excellent example of that phenomenon.

Source: https://fortune.com/2018/10/10/amazon-ai-recruitment-bias-women-sexist/

Dutch Childcare Benefits Scandal

ALGORITHM WATCH

ABOUT / PROJECTS / PUBLICATIONS / STOR

/How Dutch activists got an invasive fraud detection algorithm banned

The Dutch government has been using SyRI, a secret algorithm, to detec possible social welfare fraud. Civil rights activists have taken the matter court and managed to get public organizations to think about less repressive alternatives.

STORY 6 APRIL 2020 #AUTOMATINGSOCIETY #NETHERLANDS #PUBLICSECTOR #WELFA

POLITICO

Enter keyword

EXPLORE NEWSLETTERS & PODCASTS

Dutch scandal serves as a warning for Europe over risks of using algorithms

The Dutch tax authority ruined thousands of lives after using an algorithm to spot suspected benefits fraud – and critics say there is little stopping it from happening again.

Definitions…

Defining bias in computational & algorithmic systems

Certain aspects of a dataset are more heavily weighted and/or represented than others. This can cause low accuracy and skewed outcomes.

Different performance for diverse groups of people.

Computational systems that systematically and unfairly discriminate against certain individuals or groups of individuals in favor of others

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What are the key differences between these definitions?

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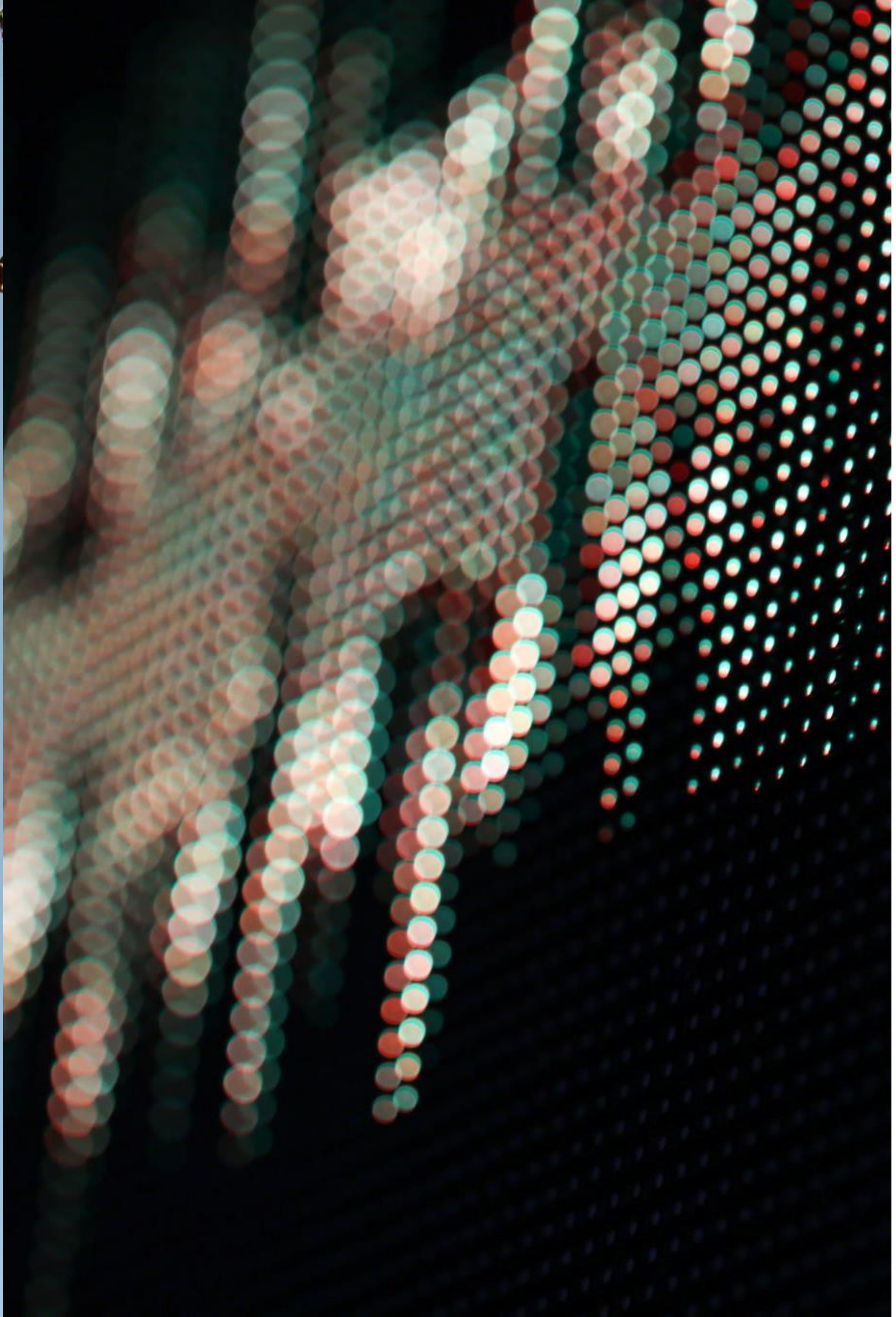

A key focus of this course…

“computer systems that systematically and unfairly discriminate against certain individuals or groups of individuals in favor of others.”

Bias in Computer Systems - Friedman & Nissenbaum, 1996

Pre-existing bias

Pre-existing bias has its roots in social institutions, practices, and attitudes.

When computer systems embody biases that exist independently, and usually prior to the creation of the system, then the system exemplifies pre-existing bias.

Pre-existing bias can enter a system either through the explicit and conscious efforts of individuals or institutions, or implicitly and unconsciously, even in spite of the best of intentions.

What are some sources or examples of pre-existing bias?

Pre-existing bias

Individual
"Bias that originates from individuals who have significant input into the design of the system, such as the client commissioning the design or the system designer (e.g., a client embeds personal racial biases into the specifications for loan approval software)".

Societal
"Bias that originates from society at large, such as from organizations (e.g., industry), institutions (e.g., legal systems), or culture at large (e.g., gender biases present in the larger society that lead to the development of educational software that overall appeals more to boys than girls)".

Technical Bias

Technical bias arises from technical constraints or technical considerations. For example:

Decontextualized Algorithms
Bias that originates from the use of an algorithm that fails to treat all groups fairly under all significant conditions

Formalization of Human Constructs
Bias that originates from attempts to make human constructs such as discourse, judgments, or intuitions amenable to computers: when we quantify the qualitative, discretize the continuous, or formalize the nonformal

Q. What is a human construct or concept used in ML tasks that might be difficult to formalise?

Emergent Bias

Emergent bias arises in a context of use with real users. Typically emerges some time after a design is completed, as a result of changing societal knowledge, population, or cultural values.

- New Societal Knowledge
- Mismatch between Users and System Design
- Different Expertise
- Different Values

Bias is multifaceted!

- There are many different taxonomies of bias
- Not expected to know all types
- Do need to understand how bias enters at different points in the pipeline, from societal factors through to data collection, preparation, model development and deployment!
- Here we will explore some specific subtypes to start to navigate this…

Check blackboard for associated reading & tasks

Friedman, B., & Nissenbaum, H. (1996). Bias in computer systems. ACM Transactions on information systems (TOIS), 14(3), 330-347.

- In the paper the explore a number of examples, in each try to identify:
- What went wrong?
- Who was harmed?
- Who benefitted?
- What (if anything) is offered as a way to mitigate such harm in the future?
- Are there any limitations to the approach?
- From1996 – is the approach appropriate for modern-day systems that incorporate machine learning?

- The Trouble with Bias by Kate Crawford (NeurIps 2017)